Floating, Caption and Equations in LaTeX

NCSC101

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November 10, 2025

What are Floating Environments?

- Objects like figures and tables are often treated as "floats."
- LaTeX automatically positions floats on the page to optimize layout.
- This prevents large figures/tables from creating awkward page breaks.
- Common floating environments:
  - `figure`
  - `table`

The `figure` Environment

```
\begin{figure}[htbp]
  \centering
  \includegraphics[width=0.8\textwidth]{example-image}
  \caption{A descriptive caption.}
  \label{fig:myfigure}
\end{figure}
```

The `figure` Environment

- `\begin{figure}` and `\end{figure}`: Enclose the figure content.
- `[htbp]`: Placement specifiers (optional, but recommended).
  - `h`: Place the figure "here" (approximately).
  - `t`: Place the figure at the "top" of a page.
  - `b`: Place the figure at the "bottom" of a page.
  - `p`: Place the figure on a separate "page" of floats.
  - `!`: Override internal parameters (use with caution). Example: `[!h]`.

  LaTeX tries these in order; you can use a subset (e.g., '[hb]').
- `\centering`: Centers the figure horizontally.
- `\includegraphics`: Includes the image file (requires the `graphicx` package). Common options: `width`, `height`, `scale`.
- `\caption`: Provides a caption for the figure.
- `\label`: Assigns a label for cross-referencing.

What are Captions?

- Captions provide brief descriptions for figures, tables, and other floating objects.
- They typically appear below the object (for figures) or above the object (for tables).
- They help readers quickly understand the content of the object without having to read lengthy explanations in the main text.
- They are essential for accessibility and clarity, especially in academic writing.

Creating Captions

- Use the **`\caption{}`** command within the environment of the floating object (e.g., **`figure`**, **`table`**).
- The text inside the curly braces becomes the caption text.

Examples

Example (Figure):

```
\begin{figure}
  \centering
  \includegraphics[width=0.5\textwidth
      ]{example-image}
  \caption{This is a sample image.}
\end{figure}
```

Example (Table):

```
\begin{table}
  \centering
  \caption{This is a
      sample table.}
  \begin{tabular}{c c}
    Header 1 & Header 2 \\
    Data 1 & Data 2 \\
  \end{tabular}
\end{table}
```

Customizing Captions

- The **`caption`** package provides many options:
  - **`\captionsetup`**: Control font size, style, label format, and more.
  - Example: **`\captionsetup{font=small,labelfont=bf}`** (sets caption font to small and label to bold).
- Even without the **`caption`** package, you can manually format captions:
  - Use font commands (e.g. **`\textbf{}`**, **`\textit{}`**, **`\small`**) inside the **`\caption{}`** command.
  - Example:
    `\caption{\textbf{Figure 1:} \textit{A description.}}`.

What are Labels?

- Labels are unique identifiers assigned to objects within your document (figures, tables, equations, sections, etc.).
- They allow you to refer to these objects later using cross-references.
- Labels are **invisible** in the compiled document.

Creating Labels

- Use the **`\label{}`** command.
- Place the **`\label{}`** command *immediately after* the **`\caption{}`** command (for figures and tables), or *after* the command that defines the object (for sections, equations, etc.).
- Choose meaningful label names. A common convention is to use prefixes:
  - **`fig:`** for figures
  - **`tab:`** for tables
  - **`eq:`** for equations
  - **`sec:`** for sections

Cross-referencing with \ref

- Use the **`\ref{}`** command to create a cross-reference to a labeled object.
- The argument of **`\ref{}`** is the label name.
- LaTeX automatically replaces **`\ref{label_name}`** with the corresponding object's number (e.g., "Figure 1", "Table 2", "Section 3").

Example:

```
As shown in Figure \ref{fig:sample_image}, the results...
```

`\ref` and `\pageref`

- `\ref` produces the number of the section, figure, table, etc.
- `\pageref` produces the *page number* where the label is located.

Example:

```
The details are provided in Section \ref{sec:methods}
on page \pageref{sec:methods}.
```

Using **hyperref** for Clickable References

- The **hyperref** package (already included at the beginning) makes references (and citations) clickable links in the PDF.
- Clicking on a reference will take you to the corresponding labeled object.
- Customize link colors with options to **hyperref**:

```
\usepackage[colorlinks=true,linkcolor=blue,citecolor=green]{
   hyperref}
```

Long Single Equations: `multline` and `multline*`

Breaking a Single **Display** Equation

Use **`multline`** (numbered) or **`multline*`** (unnumbered) for a *single* equation that is too long for one line. **This is a display environment.**

Code (`multline`)

```
A very long sum starting with $x = a+b+c+ \dots$
\begin{multline} \label{eq:long_sum}
  x = a + b + c + d + e + f + g + h + i \\
    + j + k + l + m + n + o + p + q \\
    + r + s + t + u + v + w + y + z
\end{multline}
Check \cref{eq:long_sum}.
```

Long Single Equations: `multline` and `multline*`

Breaking a Single **Display** Equation

- The text starts with an inline math snippet '$x = a + b + c + \dots$'.
- The 'multline' equation is split across three lines, formatted as described before (left, center, right alignment).
- A single equation number appears.

Expected Output

A very long sum starting with $x = a + b + c + \dots$

$$\begin{multline}
x = a + b + c + d + e + f + g + h + i \\
+ j + k + l + m + n + o + p + q \\
+ r + s + t + u + v + w + y + z \tag{1}
\end{multline}$$

Check eq. (1).

Splitting Within Environments: `split`

Aligning Parts of a Single **Display** Equation

The **`split`** environment is used *inside* another display math environment (like **`equation`**, **`align`**, **`gather`**) to break a single logical equation with alignment points (**&**). It uses the number of the surrounding **display environment**.

Code (`split` inside `equation`)

```
The Hamiltonian $H_c$ is given by:
\begin{equation} \label{eq:split_example}
\begin{split}
 H_c &= \frac{1}{2n} \sum_{l=0}^{n-1} \left( l^2 + l + \frac
     {1}{3} \right) \\
    &\quad + \frac{1}{2n} \sum_{l=0}^{n-1} \left( \dots \right)
        \\
    &= \text{Something simpler}
\end{split}
```

Splitting Within Environments: `split`

Aligning Parts of a Single **Display** Equation

- Text uses inline math 'H_c'.
- The multi-line formula appears as a single display block, aligned as specified.
- The entire block receives only one equation number.

Expected Output

The Hamiltonian H_c is given by:

$$\begin{split} H_c &= \frac{1}{2n} \sum_{l=0}^{n-1} \left(l^2 + l + \frac{1}{3} \right) \\ &\quad + \frac{1}{2n} \sum_{l=0}^{n-1} (\dots) \\ &= \text{Something simpler} \end{split} \tag{2}$$

Matrices: `pmatrix`, `bmatrix`, etc.

Typesetting Arrays **within Math Mode**

`amsmath` provides environments for common matrix delimiters. Use `&` to separate columns and `\\` to end rows. These are used *inside* math mode (e.g., within inline '$A = ...$' or display `\[ ... \]` or `equation`).

Code (Display Math Example)

```
% Displayed Matrices
\[
  \mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}
  \quad % Some space
  \mathbf{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}
\]
% Inline Matrix Example
The matrix $M = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ is
    invertible.
```

Matrices: `pmatrix`, `bmatrix`, etc.

Typesetting Arrays **within Math Mode**

Expected Output

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \mathbf{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The matrix $M = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ is invertible.

Piecewise Functions: cases

Defining Functions by Cases **within Math Mode**

The **`cases`** environment creates a structure with a large left brace. Use **`&`** to separate the function value from the condition. Use inside display math (**`\[...\]`**, **`equation`**) or inline (**`$...$`** - though less common for multi-line cases).

```
The absolute value $|x|$ is defined as
   :
\[ % Display mode is common for cases
  |x| =
  \begin{cases}
   x, & \text{if } x \ge 0 \\
   -x, & \text{if } x < 0
  \end{cases}
\]
```

```
\begin{equation} \label{
   eq:piecewise_f}
f(x) =
  \begin{cases}
   \sin(x)/x, & x \neq 0
      \\
   1, & x = 0
  \end{cases}
\end{equation}
```

Piecewise Functions: `cases`

Expected Output

The absolute value $|x|$ is defined as:

$$|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

$$f(x) = \begin{cases} \mathsf{sin}(x)/x, & x \neq 0 \\ 1, & x = 0 \end{cases} \tag{3}$$

Controlling Numbering: `\nonumber`

Suppressing Equation Numbers for Specific **Display** Lines

Within environments like **`align`** or **`gather`**, you can prevent a specific line from being numbered using **`\nonumber`** (or its synonym **`\notag`**). **This applies to lines in numbered display environments.**

```
\begin{align}
  y &= mx + c \\
  E &= mc^2 \nonumber \\ % This line won't be numbered
  F &= ma
\end{align}
```

Controlling Numbering: `\nonumber`

Suppressing Equation Numbers for Specific **Display** Lines

Expected Output

$$y = mx + c \tag{4}$$
$$E = mc^2$$
$$F = ma \tag{5}$$

Floating, Caption and Equations in LaTeX

Thank You for Listening!